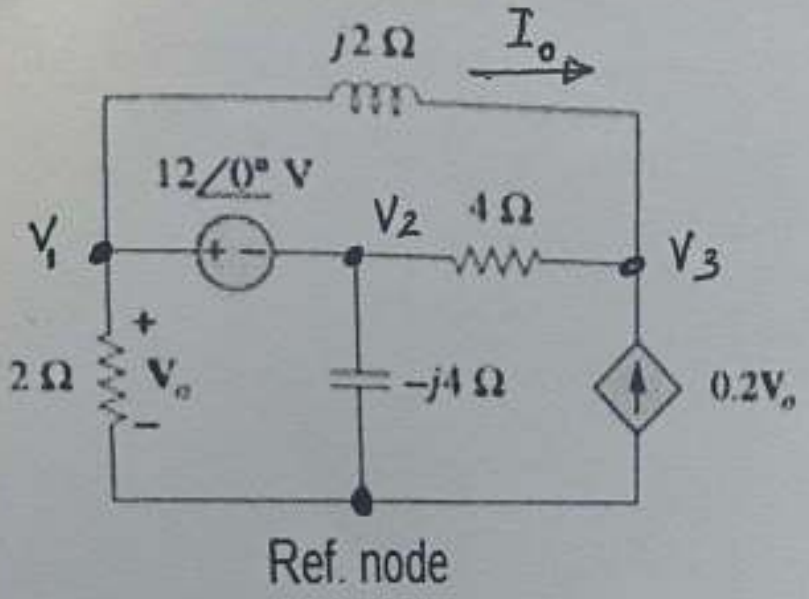


Fig.2

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10) The voltage (V0) in phasor domain is:

$V_o = V_1$

11) The current (I0) in phasor domain is:

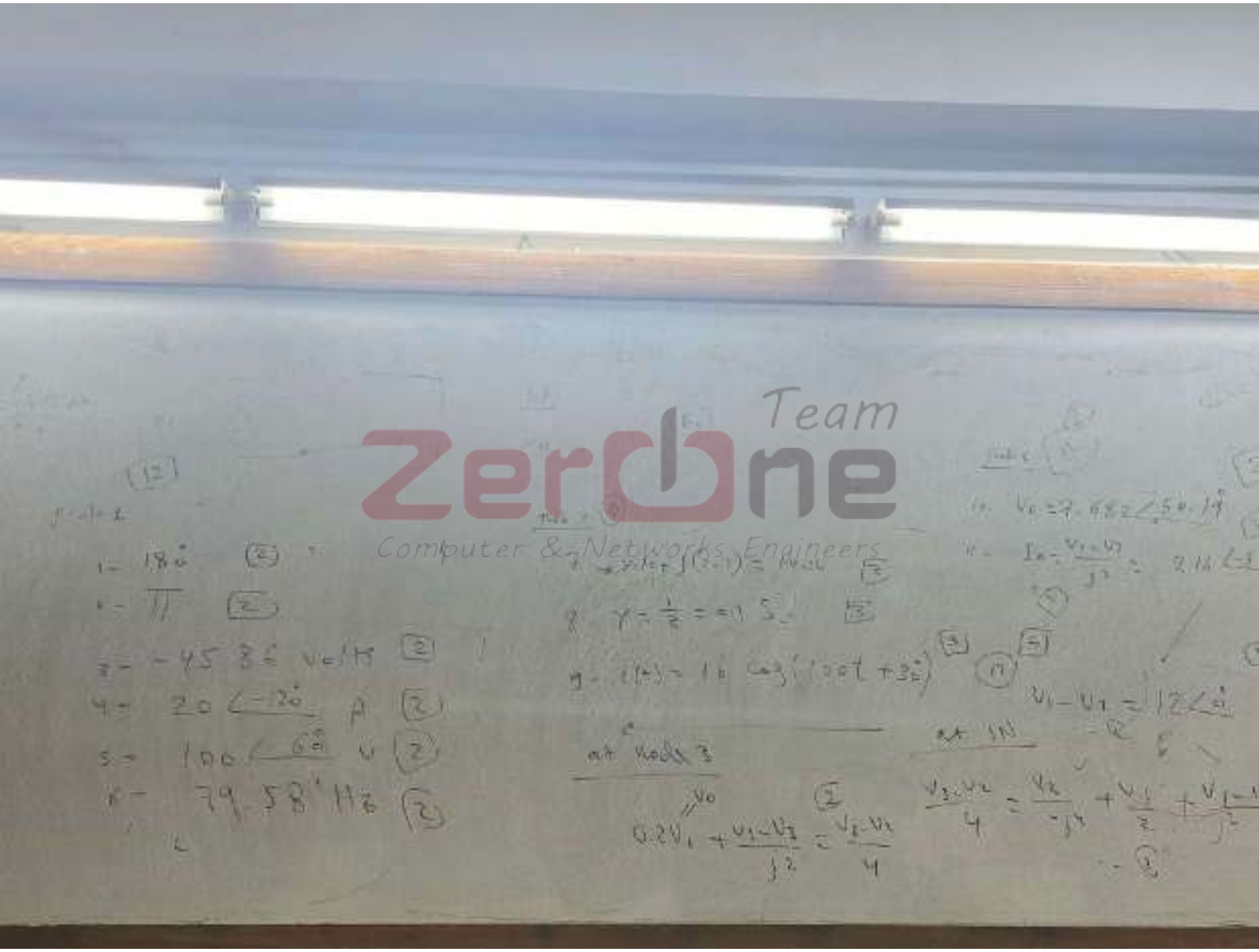
$\frac{V_1 - V_3}{2j}$

12) Write the required nodal equations (do not simplify):

$\frac{V_1}{2} + \frac{V_1 - V_3}{2j} + \frac{V_2 - V_3}{4} + \frac{V_2}{-4j} = 0$ ~(1)

$0.2V_1 = \frac{V_3 - V_2}{4} + \frac{V_3 - V_1}{2j}$ ~(2)

$V_1 - V_2 = 12$ ~(3)


Team
ZerOne
Computer & Networks Engineers

Problem 1. For the questions 1,2,3,4,5,6 consider:
$v(t) = 100\cos(500t + 60^\circ)$ and $i(t) = 20\sin(500t - 30^\circ)$.

1) The voltage leads the current in degrees by:
90

2) The current lags the voltage in radians by:
$\frac{90}{180} \times 180 = \frac{1}{2}\pi = 1.57$

3) The voltage value (v) at t=2ms is:
$100\cos(\frac{180}{\pi} \times 60^\circ) = -45.85$

4) The phasor domain representation of i(t) is:
$100\angle 60$

5) The phasor domain representation of v(t) is:
$20\angle -30^\circ$

6) The frequency of the voltage in Hz is:
$\frac{500}{2\pi} = 79.57$

Problem 2. For the questions 7,8,9 consider the circuit shown in figure 1 and answer:

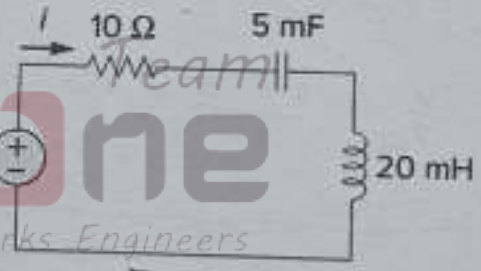


Fig. 1

- $v_s(t) = 500\cos(100t + 30^\circ)$ V

7) The equivalent impedance at the terminals of the voltage source in (Ω) is:
10

8) The admittance at the terminals of the voltage source in (S) is:
$\frac{1}{10} = 0.1$

9) The steady state current (i) in time domain is:
$I(t) = 4.7 + 2.5i = 5.3\angle 28^\circ = 5.3\cos(100t + 28^\circ)$

Problem 3. For the questions 10,11,12 consider the circuit shown in figure 2 and answer using nodal analysis: